



DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECT PROPOSAL

1. Project Title: - EmotionSync: Intelligent Feedback System using emotion detection

2. Project Scope: -

Project Overview:

This project aims to develop a system that can automatically collect and analyze user feedback based on their facial expressions. By using deep learning models to detect emotions in real-time, the system can provide insights into user sentiment and preferences. Potential applications include customer satisfaction surveys, product testing, and market research. Key challenges include handling variations in lighting, occlusion, and individual facial features, as well as ensuring ethical considerations related to privacy and bias.

Project Objectives:

- Develop a system capable of automatically detecting and analyzing facial emotions in realtime.
- Correlate detected emotions with user feedback or sentiment.
- Provide actionable insights based on the collected data.

Project Implementation:

Facial Emotion Detection Module:

- Utilize deep learning models (e.g., CNN, LSTM) for accurate facial expression recognition.
- Handle variations in lighting, occlusion, and individual facial features.

Real-time Video Processing Module:

- Capture and process video frames efficiently for real-time analysis.
- Optimize for low latency and resource consumption.

Feedback Analysis Module:

- Develop algorithms to interpret detected emotions and correlate them with user feedback.
- Consider contextual factors, emotion intensity, and temporal patterns.

Data Storage and Visualization Module:

- Implement a robust database for storing collected data.
- Provide visualization tools for analyzing trends, patterns, and insights.

Project Deliverables:

- A fully functional prototype of the automatic feedback receiver system.
- A detailed technical report documenting system architecture, algorithms, and evaluation results.
- A user manual for system operation and maintenance.

3. Requirements: -**○ Hardware Requirements**

1. PC/Laptop with a medium level computing power

○ Software Requirements

1. Machine Learning
2. CNN (Convolutional Neural Network
3. Python

STUDENTS DETAILS

Name	UID	Signature
Hari Vardhan Reddy	21BCS6250	
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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)
Priyanka Kaushik (E13618)	Supervisor	